Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/12/2025
PI Name:	Kelton, Kenneth F. Ph.D.		
Project Title:	NASA Research Under Project QUASI and ESA-Based Investigations THERMOLAB-ISS and ICOPROSOL		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	08/01/2021	**End Date:**	07/31/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Co-Investigator Anup Gangopadhyay, Ph.D. will take over as Principal Investigator (PI) for this investigation. The previous PI, Kenneth Kelton, has passed away (Ed., 4/9/26).		
COI Name (Institution):	Gangopadhyay, Anup Ph.D. (Washington University in St. Louis)		
Grant/Contract No.:	80NSSC21K1649		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

This grant is a continuation of previous grants NNX10AU19G and NNX16AB52G. Funding is provided for PI Prof. Dr. Kenneth Kelton and CoPI Dr. Anup Gangopadhyay for their role in incorporating all ground-based studies and flight preparation activities by the QUASI team at Washington University in St. Louis (WUSTL). These support QUASI flight investigations on the International Space Station (ISS) and collaborative flight projects with ESA teams ICOPROSOL (scientific coordinator – Dr. Dirk Holland-Moritz, DLR) and THERMOPROP (scientific coordinator – Prof. Dr. Hans Fecht, Ulm University). The flight experiments on the ISS are in designated batches. Batch 1 and batch 2 experiments have now been successfully completed. Batch 3 has almost been completed and Batches 4 and 5 have been approved. Metallic alloy liquids processed in a containerless environment are investigated in all of the proposed measurements for QUASI, THERMOPROP and ICOPROSOL. The primary goals of the flight and ground-based experiments are:

1. An investigation of the role of stirring on crystal nucleation from a supercooled metallic liquid.

2. Measurements of key parameters needed for a quantitative analysis of the nucleation data obtained in the ground-based and ISS studies (driving free energy, specific heat and viscosity) and for investigations of the connection between structural ordering and thermophysical properties.

3. Measurements of the electrical resistivity of metallic liquids as a function of temperature to investigate a proposed correlation between structural changes in the liquid and dynamical properties, and to confirm the existence of a plateau in the temperature dependence of the resistivity that we have recently discovered at high temperatures.

4. Measurements of the thermal conductivity of metallic liquids at high temperature.

5. Measurements of the crystal growth velocity in the ground-based and ISS studies.

6. Studies of the nucleation step in eutectic solidification.

Rationale for HRP Directed Research:

Measurements of the maximum supercooling of free-cooling metallic liquids that crystallize to phases of a different composition than the liquid, are made in the quiescent environment of the electrostatic levitation facility (WU-BESL) located in our laboratory at Washington University in St. Louis. Long-range diffusion is required for nucleation in these cases. These ground-based results are compared with undercooling studies made as a function of stirring in the ISS-EML. According to the Coupled Flux Model for nucleation, developed by Kelton, the nucleation rate should increase with the stirring rate, since long-range diffusion will become less important. A key goal of our ongoing research is to investigate this.

Measurements of the thermophysical properties, including the specific heat, density, surface tension, viscosity, thermal conductivity and electrical resistivity, will be made using the WU-BESL and the ISS-EML, since these facilities have different capabilities for these studies. The specific heat, density, surface tension and viscosity are needed for a quantitative analysis of the nucleation data. Since our earlier studies have demonstrated that ordering in the liquid couples to the nucleation barrier, the undercooling and thermophysical property measurements will be correlated with x-ray synchrotron diffraction results for the supercooled liquids of interest, obtained previously using WU-BESL. Based on our previous studies, the specific heat and electrical resistivity are also valuable for providing structural evidence for a cooperative cross-over behavior in the shear viscosity at high temperatures. When coupled with the measured electrical resistivity, measurements of the thermal conductivity will allow an investigation of the Widemann-Franz law for high temperature metallic liquids, a case where it is not been studied before.

Studies of the effects of stirring during the solidification of a eutectic Zr80Pt20 liquid have shown that nucleation proceeds by a non-Classical Nucleation Theory 2-step process, with primary nucleation of an icosahedral quasicrystal phase followed by the decomposition of this phase to the eutectic phase mixture. This raises the question of the nucleation mechanism for binary eutectic phase in general. Since eutectic solidification is important in many applications, we are investigating this process in other eutectic liquids.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Product advancement in industries such as information technology, aerospace, automotive, and basic and new materials manufacturing requires advanced materials development. In the past, this was often carried out through a laborious process of guided trial and error. The vast increases in computing power have now opened the door for realistic modeling approaches, which have the potential to greatly improve the efficiency of materials development. In the US, this has become a central focus of research in materials science. However, progress is frequently hampered by the lack of relevant thermophysical property data needed for the modeling. Further, key processes such as crystal nucleation from the liquid or other phase transitions within the liquid, are often poorly understood, making it difficult to realistically model them. In many cases, the experimental data and a deeper understanding of some processes, cannot be obtained from terrestrial experiments due to interactions with sample containers at the temperatures of interest. The combination of the containerless based methods of WU-BESL and the ISS-EML enables these measurements. In addition to their relevance for processes such as additive manufacturing on earth, where nucleation in highly stirred conditions is important, measurements in two different gravitational fields will be important for designing additive manufacturing in space and on the Moon. Finally, past measurements of the electrical resistivity in high temperature liquids have revealed a plateau in the temperature dependence of the resistivity at temperatures near a crossover temperature for the viscosity. This is not predicted by any existing theory. Our measurements will provide valuable information for developing a theoretical understanding of this phenomenon.

APRIL 2026 UPDATE:
This project continues under a new Principal Investigator: Anup Gangopadhyay, Ph.D. For current reporting, see Grant 80NSSC21K1649 under Anup Gangopadhyay, Ph.D. (Ed., 4/9/26).

A summary of activities caried out during year 4 of this grant are listed below.

1. Measured thermophysical and electrical properties on the International Space Station (ISS) in August 2024 on the Batch 3 sample Zr80Pt20 (ID#70).

Task Progress:

2. Attempted measurements on the ISS of Ti45Zr45Ni10 (ID#85), for which QUASI is a Co-PI, in 2024. Unfortunately, these were abandoned after several attempts to activate the sample transfer mechanism failed. Because of this, ICOPROSOL and QUASI requested that the international working group (IWG) committee allow this composition to be included in Batch 5 for further investigations. This request was granted.

3. Completed the analysis of the nucleation kinetics as a function of stirring, using our ground-based electrostatic levitation (ESL) and the ISS electromagnetic levitatator (EML), of Zr80Pt20 (ID#70). Contrary to expectation, the dynamical pre-factor for nucleation decreased with increased stirring. This was shown to be due to heterogeneous nucleation on the surface of the liquid drop. Stirring likely moved the catalytic sites into the hotter interior of the liquid making them less effective as heterogeneous sites. The nature of the heterogeneities is not yet known, but may be due to ordered patches on the surface of the liquid, as has been observed in other liquids. The results were published in the Journal of Chemical Physics in 2025.

4. Analyzed experimental data from the ISS-EML experiments for the surface tension, and viscosity, of Zr80Pt20 (ID#70) with collaborators Stephan Schneider and Mitja Becker of The Microgravity User Support Center (MUSC), Deutsches Zentrum für Luft- und Raumfahrt (DLR), Köln. The measured surface tension decreases with increasing temperatures as expected and fits well with the Butler equation. The viscosities measured as a function of temperature in the ISS-EML and ground-based ESL show some differences, suggesting an effect of fluid flow and levitation strength on the measurements.

5. Finished the analysis of the specific heat data obtained on the ISS-EML for the glass, crystal and equilibrium and supercooled liquids of Cu47Zr47Al6 (ID# 41). A manuscript discussing this was submitted to Acta Materialia in March 2025. Another manuscript on a comparative study of the specific heats of quasicrystal forming Ti39.5Zr39.5Ni21 and Ti45Zr45Ni10 alloys in the crystal and liquid states is in preparation.

6. Finished the analysis of the kinetics of crystal nucleation from a Zr47Cu47Al6 liquid that were measured in the ground-based ESL and with the ISS-EML. Unfortunately, as mentioned earlier, the statistics were insufficient in the ISS studies to draw firm conclusions on the effect of stirring. More studies are planned in Batch 4.

7. Continued molecular dynamics studies of crystal nucleation in Zr47Cu47Al6 liquids to support the experimental results. The accuracy of the potential was confirmed by a comparison of the calculated structure factures, S(q), with our experimental results from X-ray synchrotron scattering measurements, as well as a comparison between the calculated and measured liquidus temperatures. Seeding calculations were made to determine the interfacial free energy for the nucleus. The results were inconsistent with experimental studies, reflecting a failure of the commonly used Classical Nucleation Theory (CNT) to account for the diffuse interface of the nucleating cluster.

8. While growth of eutectic phases has been well studied, there are very few experimental studies of the nucleation step. From synchrotron scattering data from levitated liquids using our ground-based ESL, we have analyzed the primary phase formation during eutectic phase nucleation for many eutectic liquids, including Zr80Pt20, Zr82Ir18, Zr77Rh23, Zr75.5Pd24.5, Zr54Cu46, Zr44Cu56, and Zr36Ni64. The nucleation mechanism varies. In some cases, nucleation proceeds by a non-CNT two step nucleation processing, first nucleating a metastable phase that subsequently transforms to the equilibrium eutectic phase mixture. In other cases, however, both eutectic phases nucleated directly from the liquid. A goal for future studies is to identify the reasons for this different behavior.

9. Planned activities for June and July 2025 are:

a. Continue to work on the preparation of two manuscripts (listed below),

(i) A. K. Gangopadhyay, M. Becker, S. Schneider, and K. F. Kelton, “Surface Tension and viscosity of Zr80Pt20 measured with ESL on earth and under microgravity with EML on the International Space Station”.

(ii) A. K. Gangopadhyay, D. Holland-Moritz, and K. F. Kelton, “A comparative study of the specific heats of quasicrystal forming Ti39.5Zr39.5Ni21 and Ti45Zr45Ni10 from terrestrial and microgravity measurements aboard the International Space Station.”

b. Prepare five 6.5mm diameter samples of composition, Ti42Zr37Ni21, for electrical coupling measurements and send them to Microgravity Users Support Center (MUSC), DLR, Koln, Germany. This composition has been selected as one of the Batch 5 samples for experiments on the ISS-EML.

c. Dr. Anup Gangopadhyay will attend the IWG#24 meeting in Madrid on June 16 and 17 and present the most recent results from experiments on the ISS.

d. Continue to analyze the surface tension and viscosity data for Cu47Zr47Al6, (ID# 41) which were collected during Batch 3 experiments and start preparing a manuscript for publication.

e. Begin work on a manuscript on Bulk Modulus estimations for a few glass forming liquids for which specific heat and thermal expansion measurements have been performed.

f. Continue work on the molecular dynamics of nucleation (MD) in Cu47Zr47Al6, beginning to incorporate the Diffuse Interface Theory of nucleation to account for the diffuse interface found in the MD calculations.

g. Continue the analysis of the synchrotron diffraction data for crystallization of the metallic eutectic liquids.

10. Planned studies in the next year are:

a. Finalize the three manuscripts listed above (point 9) and submit them for publication.

b. Complete the molecular dynamics studies of nucleation (MD) in Cu47Zr47Al6 and prepare and submit a manuscript for publication.

c. Complete the analysis of the synchrotron diffraction data for crystallization of the metallic eutectic liquids and prepare and submit a manuscript for publication.

d. Begin process studies on the ISS for the remaining studies in Batch 3 and beginning studies in Batch 4.

e. The preparation of the PhD thesis of the graduate student working on this grant, with a defense of the work in late spring or early summer 2026, followed by her graduation in May 2026 or August 2026.

	These accomplishments and planned activities are described in greater detail in the annual technical document. [Ed. Note: a technical progress update is delivered annually to NASA.]
Bibliography Type:	Description: (Last Updated: 05/20/2025)
Articles in Other Journals or Periodicals	Gangopadhyay A, Kelton K. "Temperature dependent specific heat measurements of equilibrium, supercooled liquid, glass, and the corresponding crystal of Cu47Zr47Al6 from terrestrial and microgravity experiments on the International Space Station." Acta Materialia, under review as of May 2025. http://dx.doi.org/10.2139/ssrn.5221535 , May-2025
Articles in Other Journals or Periodicals	Gangopadhyay AK, Holland-Moritz D, Kelton KF. "A comparative study of the specific heats of quasicrystal forming Ti39.5Zr39.5Ni21 and Ti45Zr45Ni10 from terrestrial and microgravity measurements aboard the International Space Station." Manuscript in preparation as of May 2025. , May-2025
Articles in Other Journals or Periodicals	Gangopadhyay AK, Becker M, Schneider S, Kelton KF. "Surface tension and viscosity of Zr80Pt20 measured with ESL on Earth and under microgravity with EML on the International Space Station." Manuscript in preparation as of May 2025. , May-2025
Articles in Other Journals or Periodicals	Sheng Y, Kelton KF. "Identification of the nucleation mechanism for eutectic solidification - X-ray synchrotron studies of containerless-processed metallic liquids." Manuscript in preparation as of May 2025. , May-2025
Articles in Peer-reviewed Journals	Gangopadhyay AK, Sheng Y, Bracker GP, Hyers RW, Kelton KF. "Slower nucleation kinetics with stirring in a supercooled Zr80Pt20 liquid—Terrestrial and microgravity experiments on the International Space Station." Journal of Chemical Physics. 2025 Apr 7;162(13). https://doi.org/10.1063/5.0263986 , Apr-2025
Conference Materials (Downloadable)	Kelton KF, Gangopadhyay AK, Sheng Y, Bracker GP, Hyers RW. "Influence of stirring on the nucleation kinetics in supercooled metallic liquids." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Jan-2025 Kelton_ASGSR_Presentation_12_2024.pptx